

- 1 A child kicks a ball so that it leaves horizontal ground with a velocity of 28 m s^{-1} at an angle of 34° to the horizontal, as shown in Fig. 1.1.

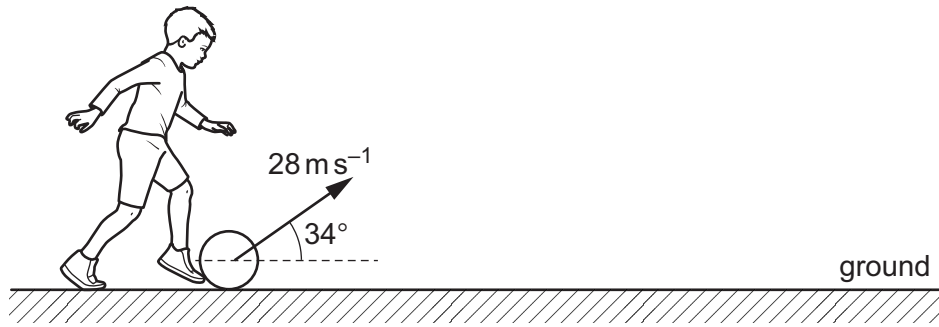


Fig. 1.1

Air resistance is negligible. The ball leaves the ground at time $t = 0$.

- (a) (i) Calculate the horizontal component v_H and the vertical component v_V of the velocity of the ball immediately after it has left the ground.

$$v_H = \dots\dots\dots \text{ m s}^{-1}$$

$$v_V = \dots\dots\dots \text{ m s}^{-1}$$

[2]

- (ii) Show that the ball reaches its maximum height at time $t = 1.6 \text{ s}$.

[1]



- (iii) On Fig. 1.2, sketch the variation of v_H with time t between $t = 0$ and $t = 3.2$ s. Label your line H.

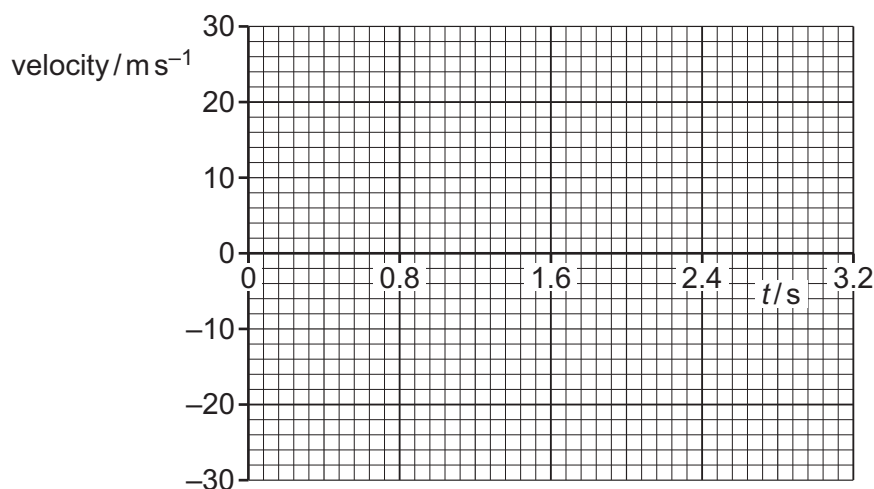


Fig. 1.2

[1]

- (iv) On Fig. 1.2, sketch the variation of v_V with time t between $t = 0$ and $t = 3.2$ s. Assume that velocity in the upward direction is positive. Label your line V. [3]
- (b) The total change in momentum of the ball between leaving the ground at $t = 0$ and landing on the ground at $t = 3.2$ s is 13 kg ms^{-1} .

- (i) Define momentum.

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 [1]

- (ii) Calculate the force that acts on the ball while it is in the air.

force = N [2]

- (iii) Determine the mass of the ball.

mass = kg [1]

[Total: 11]

[Turn over]

